Standard Operating Procedures for running Demultiplexing in linux system

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1 Objective

This document is generated to show the Bioinformatician how to do the Demultiplexing on the linux operating system.

2 Scopes

This document is used for the Bioinformatics team.

3 Responsibilities

3.1 The Bioinformatics officer is responsible for the preparation of this SOP；

3.2 The Bioinformatics leader is responsible for reviewing the standard operating instructions and ensuring they are newly updated and effective；

3.3 Laboratory director is responsible for approving this SOP；

3.4 Bioinformatics analysts are responsible for doing analysis in accordance with this SOP.

4 Terms and definitions

NA

5 Operating Procedure/Quality standards/Management requirements

5.1 Install the software

Before operating, the analyst should download either of two software, depending on the WFS software they are using. The software and manual are available on the Microsoft Sharepoint:
the version for WFQ version-basecallLite :
https://bgitech.sharepoint.com/teams/MGI_FBS_TeamAPAC/Shared%20Documents/General/02_Frequently_used_softwares/02_Demultiplexing/splitBarcode_V2.0.0_release_4_basecallLite_MGI.zip
the version for WFQ version-zebracall:
https://bgitech.sharepoint.com/teams/MGI_FBS_TeamAPAC/Shared%20Documents/General/02_Frequently_used_softwares/02_Demultiplexing/splitBarcode_V2.0.0_release_4_zebracall_BGIR.zip
the manual:
https://bgitech.sharepoint.com/teams/MGI_FBS_TeamAPAC/Shared%20Documents/General/02_Frequently_used_softwares/02_Demultiplexing/splitBarcode_manual.pdf

Note: some parameters between demultiplexing 1.6 and 2.0 are a bit difference, please refer to manuals for more information.

5.2 prepare barcode and check FASTQ files

1) prepare barcode files.

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The demultiplexing accepts only one barcode file with name and sequence included. The format of the file should be like Fig 1:

```
192.168.233.12 - PuTTY
2_2      GGACGGAATCGGACGGAATC
3_3      CTTACTGCCGCTTACTGCCG
4_4      ACCTAATTGAACCTAATTGA
5_5      TTCGTATCCGTTCGTATCCG
6_6      GGTAACGAGCGGTAACGAGC
7_7      CAACGTATAACAACGTATAA
8_8      ACGTCGCGTTACGTCGCGTT
9_9      TTCTGCTAGCTTCTGCTAGC
10_10    AGGAAGATAGAGGAAGATAG
11_11    GCTCTTGCTTGCTCTTGCTT
12_12    CAAGCACGCACAAGCACGCA
13_13    CGGCAATCCGCGGCAATCCG
```

Fig 1. An example of barcode file.

To better use the software, here are some tips for preparing the barcode files:

- ID and sequence are tab delimited
- No blank line
- Only English letter and number for ID
- ID or sequence cannot be blank
- The length of barcode sequence needs to be longer than 2nt
- For barcode sequence, only A, T, C, G or N are accepted
- In one barcode file, ID and sequence should be unique (shouldn't have the same sequence in 2 ID, or 1 ID has 2 different sequences)
- For better formatting, it is recommended to open/edit your barcode in notepad++ (instead of excel)

2) Check FASTQ files.

The single end (SE) or paired end (PE) reads should include the barcode information. So the length of the reads should be longer than expected. For example, PE100 reads with 10 bases barcode at the end of read2 should have length as 100bp + 110bp. The command line of doing this could be like Fig2.

```
(base) tianwei@login-1-1:Thu Sep 30 15:10:21 /share/Data01/tianwei/FBS/Test/Demultiplexing_test/case1_10r
$less case1_10r_1.fq.gz | head -n 2 | tail -n 1 | wc -L
100
(base) tianwei@login-1-1:Thu Sep 30 15:10:33 /share/Data01/tianwei/FBS/Test/Demultiplexing_test/case1_10r
$less case1_10r_2.fq.gz | head -n 2 | tail -n 1 | wc -L
110
(base) tianwei@login-1-1:Thu Sep 30 15:10:40 /share/Data01/tianwei/FBS/Test/Demultiplexing_test/case1_10r
$
```

Fig2. Calculate the length of paired reads

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Note: normally the unbarcoded reads from MGI sequencing platform named like: Vxxxx_read_1.fq.gz and Vxxxx_read_2.fq.gz

To check if the barcode sequence is correctly located in the reads file, the analyst should open the file and identify the barcode location, by using search of 1 barcode sequence, remember to change barcode strand if the reversed barcodes are used. Take the above case as an example, we are expecting the 10 bases barcode at the end of read2 (Fig3).

```
192.168.233.12 - PuTTY
(base) tianwei@login-1-1:Thu Sep 30 15:23:12 /share/Data01/tianwei/FBS/Test/Demultiplexing_test/case1_10r
$more case1.barcode
barcode1        TGGGAACGCT
barcode2        TAATCCTTAT
barcode3        CAATGCGTAG
barcode4        ATCCGCTCAG
barcode5        GTCTCTAATC
(base) tianwei@login-1-1:Thu Sep 30 15:23:16 /share/Data01/tianwei/FBS/Test/Demultiplexing_test/case1_10r
$cut -f2 case1.barcode | perl -ne 'chomp;$_=~tr/ATCG/TAGC/;$_=reverse($_);print $_,"\n";'
AGCGTTCCCA
ATAAGGATTA
CTACGCATTG
CTGAGCGGAT
GATTAGAGAC
(base) tianwei@login-1-1:Thu Sep 30 15:23:20 /share/Data01/tianwei/FBS/Test/Demultiplexing_test/case1_10r
$less -S case1_10r_2.fq.gz
CTCTCATGTTTTCTGTCCCCCAGGAGTGTGGTCGTCCTGCTCCTGCTAGTATTCTGCTTCTGCTTCCTGGTGGCCTGTGTCCTGTACCTGCACTTCACCCAGCGTTCCCA
+
?F?EFD1EF>@FF=BC;BBE<D5G0@,F/96G10FF8B@8DG3EC33>F6DEF=?G?'*?8F<789FF2A3E*E*1+.2DF48C4:0D31-56)*'G/B9FFFFFFFFFF
@V350022662L3C001R0020000017/2
GTGATTTTGCTTTTGTGATTCTATTACTTAGGAGATGCCAAAATTGGCAATAACAGTGTCAGCTCTTTAAAGAATGATGACTTCACCATGAGAGGTGTGCAGCGTTCCCA
+
E=FEFFFFFFFFFFFFF8FFFFFFEFFFFGFF*FFFGFEFF;FFFDCFFFFFFFFFEFFFFF4FFAFCGFF@0F@EFFFC8BD6FFFFDF7D:FDA'FF6FFFFFFFFFF
@V350022662L3C001R0020000042/2
CTGGAAATCAGGAGCACCAAGGGCAGAAGAAGATCAATGCTCCAGCTTAAGCAGTCAGGAGGAGAGCAAATTCAACCTTCTTTCACCATTTTGTTTCCTTAGCGTTCCCA
+
DFEB0CGGEF@DEFAGFFE@FFFGFFEACEBGFBGFG>;GGFAFGCFFDAGD8EF?GE3ACDGFFBDFCD<EGC=FEFGFFDFGAGFFG@CGEEFEEFEFFFFFFFFFFF
@V350022662L3C001R0020000052/2
GGACCCCTCACCTGCTTCTTCTGGTTCATTCTCCAAATCCCCGACCACCGCCGGCTGATGCTGGGCCCCGAGGTGGTGCAGGCACTGCTGGTGGGGGAGCAGCGTTCCCA
+
FFEF>FFDFFFFFFFFFFFFFFFFFFFEFFFFFGEFFDFGGFFB'FFFFGFGFFFDF>FF>FF&GFGFGG/FF0GF;F7FG?=<@=FF<FF9FF'FF2F2FFFFFFFFFF
@V350022662L3C001R0020000057/2
TGCCTCCAGGTGCCTGTGCCCATGTTCTTGCCCACTACCTTGGAGAGCACAGACAAGATTGTAGAGACCATTGAGGAGCTGAAGGTGAAGATCCCTTCCAAGCGTTCCCA
+
GFCFFFFFFFFGFFFGDGFFFEFFDFEFGFBGFGFDGFFGEGC1F:FFGBEDDBGGFAFGF9GG8G7EFF9FF9GF6FDFC1FFGBF5FG7>FFFEF=FFFFFFFFFFFF
```

Fig3. Identify the barcode location in read2

5.3 Start the demultiplexing

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Besides the reads file (**-1** and **-2**) and barcode file (**-B**), there are 2 more key parameters are required, **-b** shows the position and mismatches, and **-r** means reversed barcodes are used. The basic useage of the software is shown as Fig4.

```
code_V2.0.0_release/linux/bin/splitBarcode --help
Support cal2fq, splitBarcode
Usage: /share/Data01/tianwei/ForLab/Bin/DeMultiplexing/splitBarcode_V2.0.0_release_4_basecallLite_MGI/splitBarcode_V2.0.0_release/linux/bin/splitBarcode [OPTIONS]

Options:
  -h,--help                   Print this help message and exit
  -B TEXT:FILE                Barcode file
  -1 TEXT:FILE                Fastq file
  -2 TEXT:FILE                Fastq file for PE mode
  -o,-O TEXT                  Output dir for decoded fastq
  -b INT x 3                  Barcode information: startCycle length mismatchNum
  -t,-T INT:POSITIVE          Set the maximum thread numbers
  -m,-M FLOAT:POSITIVE        Set the maximum memory(GB)
  -f,-F TEXT:DIR              The path storing cal files
  -C INT:POSITIVE             The cycle number
  --Col INT:POSITIVE          The max column of fov lists
  --Row INT:POSITIVE          The max row of fov lists
  -n,-N TEXT                  Set slide name
  -u,-U INT:POSITIVE          Set lane number
  -s,-S                       Set single fov mode
  -p,-P INT:POSITIVE          Set read1 lenght in PE mode
  -e,-E INT:INT in [0 - 3]    Set end cycle process mode
  --start_col INT:POSITIVE    Set start column number
  --start_row INT:POSITIVE    Set start row number
  --read_len INT:POSITIVE     Set read length
  --cpu                       Set cpu mode,Default:false
  --no_filter                 Set no filtering,Default:false
  --filter_param FLOAT x 7    filter parameters information
  --saveNotFiltered           Set save unfiltered reads,Default:not save
  --barcode_start INT:POSITIVE
                              set barcode start pos.Default: -1
  --barcode_len INT:POSITIVE  set barcode length.Default: 0
  --barcode_mismatch INT:POSITIVE
                              set barcode mismatch number. Default: 1
  --barcode2_start INT:POSITIVE
                              set barcode start pos.Default: -1
  --barcode2_len INT:POSITIVE set barcode length.Default: 0
  --barcode2_mismatch INT:POSITIVE
                              set barcode mismatch number. Default: 1
  -i INT x 3                  Barcode information: startCycle length mismatchNum
  -j,-J TEXT                  the json file
  -r,-R                       Apply reverse complement of barcode sequence
  --umi                       Add barcode uid into fastq files

(base) tianwei@login-1-1:Thu Sep 30 15:33:28 /share/Data01/tianwei/FBS/Test/Demultiplexing_test/case1_10r
$
```

Fig4. Parameters for demultiplexing 2.0

For example, SE100 reads with 10 bases reversed barcode at the end of the read file (1 mismatch allowed) should have parameters like:

```
./bin/splitBarcode -B BarcodeV2.1.txt -1 test_read.fq.gz -o result_se_single_index -b 100 10 1
```

While PE100 reads with 10 bases reversed barcode at the end of read2 (1 mismatch allowed) should have parameters like:

```
./bin/splitBarcode -B BarcodeV2.1.txt -1 test_read_1.fq.gz -2 test_read_2.fq.gz -o result_pe_single_index -b 200 10 1 -r
```

5.4 Check output files

The output of the demultiplexing contains FASTQ files for each barcode and some statistics files(Fig 5).

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```
$ll
total 258M
-rw-r--r-- 1 bcbuild ST_BI 1.6K 2020/03/17 11:24 BarcodeStat.txt
-rw-r--r-- 1 bcbuild ST_BI  11K 2020/03/17 11:24 G10000955E320oldDyeB_L01_100_1.fq.fqStat.txt
-rw-r--r-- 1 bcbuild ST_BI  240 2020/03/17 11:24 G10000955E320oldDyeB_L01_100_1.fq.gz
-rw-r--r-- 1 bcbuild ST_BI  12K 2020/03/17 11:24 G10000955E320oldDyeB_L01_100_2.fq.fqStat.txt
-rw-r--r-- 1 bcbuild ST_BI  247 2020/03/17 11:24 G10000955E320oldDyeB_L01_100_2.fq.gz
-rw-r--r-- 1 bcbuild ST_BI  330 2020/03/17 11:24 G10000955E320oldDyeB_L01_101_1.fq.fqStat.txt
-rw-r--r-- 1 bcbuild ST_BI   31 2020/03/17 11:24 G10000955E320oldDyeB_L01_101_1.fq.gz
-rw-r--r-- 1 bcbuild ST_BI  330 2020/03/17 11:24 G10000955E320oldDyeB_L01_101_2.fq.fqStat.txt
-rw-r--r-- 1 bcbuild ST_BI   31 2020/03/17 11:24 G10000955E320oldDyeB_L01_101_2.fq.gz
-rw-r--r-- 1 bcbuild ST_BI  24K 2020/03/17 11:24 G10000955E320oldDyeB_L01_10_1.fq.fqStat.txt
-rw-r--r-- 1 bcbuild ST_BI  33M 2020/03/17 11:24 G10000955E320oldDyeB_L01_10_1.fq.gz
-rw-r--r-- 1 bcbuild ST_BI  330 2020/03/17 11:24 G10000955E320oldDyeB_L01_999_1.fq.fqStat.txt
-rw-r--r-- 1 bcbuild ST_BI   31 2020/03/17 11:24 G10000955E320oldDyeB_L01_999_1.fq.gz
-rw-r--r-- 1 bcbuild ST_BI  330 2020/03/17 11:24 G10000955E320oldDyeB_L01_999_2.fq.fqStat.txt
-rw-r--r-- 1 bcbuild ST_BI   31 2020/03/17 11:24 G10000955E320oldDyeB_L01_999_2.fq.gz
-rw-r--r-- 1 bcbuild ST_BI  23K 2020/03/17 11:24 G10000955E320oldDyeB_L01_undecoded_1.fq.fqStat
.txt
-rw-r--r-- 1 bcbuild ST_BI 8.0M 2020/03/17 11:24 G10000955E320oldDyeB_L01_undecoded_1.fq.gz
-rw-r--r-- 1 bcbuild ST_BI  27K 2020/03/17 11:24 G10000955E320oldDyeB_L01_undecoded_2.fq.fqStat
.txt
-rw-r--r-- 1 bcbuild ST_BI 9.6M 2020/03/17 11:24 G10000955E320oldDyeB_L01_undecoded_2.fq.gz
-rw-r--r-- 1 bcbuild ST_BI 790K 2020/03/17 11:24 SequenceStat.txt
```

Fig5. Output files

In most cases, the statistics files are very important because they give the split rate for each barcode and it could help the analyst to see if the demultiplexing has been succeeded. The files are:

a. Barcode split file

File name: *BarcodeStat.txt*

This file contains result of demultiplexing. One key piece of informationforassessing sequencing data is the ratio of the number of reads assigned toa barcode to total count, shown in last column in percentage. Each column represents:

- Barcode ID
- Number of reads assigned to a barcode group without any mismatch
- Number of reads assigned to a barcode group with mismatches detected
- All reads demultiplexed (sum of column 2 and3)
- Ratio of demultiplexed reads and total reads in percentage.

Last row is the sum for all demultiplexed barcode. The total rate is 93.61% (Fig6).

Note: Ambiguous reads are not included.

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```
$cat BarcodeStat.txt
#Barcode          Correct          Corrected        Total    Percentage(%)
barcode3          0        1        1        0.000071
barcode4          1        0        1        0.000071
barcode5          9        5        14       0.000994
barcode6          22       2        24       0.001704
barcode7          13       1        14       0.000994
barcode8          33       2        35       0.002486
barcode9          176714   7690     184404   13.096159
barcode10         356343   18274    374617   26.604866
barcode11         243836   9191     253027   17.969685
barcode12         41836    1721     43557    3.093368
barcode13         63707    3750     67457    4.790718
barcode14         61122    3336     64458    4.577733
barcode15         69034    3662     72696    5.162786
barcode16         41684    1540     43224    3.069719
barcode17         58936    2043     60979    4.330658
barcode18         29773    2802     32575    2.313439
barcode19         25431    1739     27170    1.929582
barcode20         64261    3525     67786    4.814083
barcode21         8593     394      8987     0.638246
barcode22         8428     982      9410     0.668287
barcode23         6977     712      7689     0.546064
```

```
barcode107        0        2        2        0.000142
barcode113        0        1        1        0.000071
barcode117        0        8        8        0.000568
barcode124        0        3        3        0.000213
Total     1256753          61409    1318162          93.614342
```

Fig6. An example of BarcodeStat.txt

b. Barcodestat file

File name: *SequenceStat.txt*

This file recodes statistics of barcode. If demultiplexing rate is abnormal, this file allows visual examination of distribution of barcode counts. Should incorrect barcodes were used or reverse-complementation parameter was omitted, this file offers diagnostic signs for such mistakes. Content of four columns are as following (Fig7.):

- Barcode sequence
- Barcode ID
- Count of occurrence
- Percentage in terms of total reads count

Note: In most cases the analyst should check if the most acquired sequence in the barcode position is one of the barcode sources, if not (marked as *undecoded* in the 2nd column), there could be some issues happened.

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```
$cat SequenceStat.txt | head  -20
#Sequence        Barcode         Count    Percentage(%)
CTATCTTCCT       barcode10       356343   25.307068
AAGCAAGAGC       barcode11       243836   17.316951
GCTAGCAGAA       barcode9        176714   12.550024
TCTGGAATGA       barcode15       69034    4.902715
TTGTACCAAT       barcode20       64261    4.563742
CGGATTGCCG       barcode13       63707    4.524397
GAATCCTGAT       barcode14       61122    4.340814
CATCACTCAC       barcode17       58936    4.185567
TGCGTGCTTG       barcode12       41836    2.971144
ATCCAGCATC       barcode16       41684    2.960349
CAGCTGACTC       barcode18       29773    2.114444
TTCGCAGACA       barcode19       25431    1.806080
ACCACAATCG       barcode21       8593     0.610265
GGAAGTCTGT       barcode22       8428     0.598547
AGAGTGTGGA       barcode23       6977     0.495498
CTATATTCCT       barcode10       6332     0.449691
ATATAATAAT       undecoded       3088     0.219306
GCTAGCAGGA       barcode9        2545     0.180743
ATATAATCAT       undecoded       2189     0.155460
```

Fig7. An example of SequenceStat.txt

c. FASTQ stat file

File name: *slide_lane_barcodeID.fq.fqStat.txt*

One *fq.fqStat.txt* file will be generated for each barcode (slide and lane are placeholders that will be replaced with actual slide ID and lane number). This file contains statistics for demultiplexing, for instance, read count, GC content, Q30 score and base counts (Fig8).

```
#Name    result_pe_single_index/G10000955E320oldDyeB_L01_undecoded_1.fq.gz
#PhredQual      33
#ReadNum        89915
#row_readLen    100
#col    48
#BaseNum        8991500
#N_Count        2829    0.031463
#GC%    39.40
#Q10%   73.91
#Q20%   58.60
#Q30%   44.49
#EstErr%        7.095463
#Pos    A       C       G       T       N       0       1       2       3       4       5       6
1       27147   23422   14924   24421   1       1       0       0       0       3057    13389   2539
2       37531   15875   20338   16166   5       5       0       0       0       123     1798    2049
3       36439   18157   17582   17730   7       7       0       0       0       237     1994    2118
4       25640   18224   27556   18488   7       7       0       0       0       3741    10989   3497
5       37779   15017   23030   14080   9       9       0       0       0       460     1592    2160
6       11151   27311   14262   37179   12      12      0       0       0       1546    3570    8631
7       16088   23902   15234   34688   3       3       0       0       0       2732    9605    4491
8       34236   21234   18415   16023   7       7       0       0       0       313     1677    3308
9       34231   18586   15368   21724   6       6       0       0       0       190     2730    2403
10      34924   16397   17267   21281   46      46      0       0       0       447     5379    5860
11      36843   16005   17699   19291   77      77      0       0       0       188     1873    2119
12      34914   16450   17442   21095   14      14      0       0       0       2766    2090    2680
13      33579   17411   15654   23257   14      14      0       0       0       218     6252    6618
14      31384   18103   15668   24748   12      12      0       0       0       504     7130    6798
15      33676   17635   15239   23360   5       5       0       0       0       975     4919    11139
```

Fig 8. FASTQ stat file

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5.5 Example of different cases

1. 10 bases single barcode for PE100 reads. Reversed.
   Test file:
   https://bgitech.sharepoint.com/teams/MGI_FBS_TeamAPAC/Shared%20Documents/General/01_Demo_Data_And_Reports/02_BIT_Demo_Data/05_Demultiplexing/case1_10r

   Barcode file:
   ```
   barcode1      TGGGAACGCT
   barcode2      TAATCCTTAT
   barcode3      CAATGCGTAG
   barcode4      ATCCGCTCAG
   barcode5      GTCTCTAATC
   ```

   Parameters:
   ```
   ./bin/splitBarcode -B case1.barcode -1 case1_10r_1.fq.gz -2 case1_10r_2.fq.gz -b 200 10 1
   -r -o ./
   ```

2. Dual (10bp+10bp) barcode for PE100 reads. Reversed.
   Test file:
   https://bgitech.sharepoint.com/teams/MGI_FBS_TeamAPAC/Shared%20Documents/General/01_Demo_Data_And_Reports/02_BIT_Demo_Data/05_Demultiplexing/case2_10r_10r

   Barcode file:
   ```
   barcode1      TGGGAACGCTCCGAGACGTT
   barcode2      TAATCCTTATCATTCCTATT
   barcode3      CAATGCGTAGGTAGCGTCGG
   barcode4      ATCCGCTCAGCCTGTAGGAG
   barcode5      GTCTCTAATCTTCGACCTAG
   ```

   Note: the red one stands for the 2nd barcode

   Parameters:
   ```
   ./bin/splitBarcode -B case2.barcode -1 case2_10r_10r_1.fq.gz -2 case2_10r_10r_2.fq.gz -b
   200 10 1 -b 210 10 1 -r -o ./
   ```

   Note: Please note that the software recognizes barcode by position, so if you have reads include dual barcode, for example the barcode region in read2 is **AGCGTTCCCAAACGTCTCGG**, And the original barcodes are **TGGGAACGCT** and **CCGAGACGTT**. The analyst should **NOT** prepare the barcode as whole reversing (**CCGAGACGTTTGGGAACGCT**), instead, the 2 barcodes should be prepared separately, which is **TGGGAACGCTCCGAGACGTT**.

3. Dual (10bp+6bp) barcode for PE100 reads. Reversed.
   Test file:
   https://bgitech.sharepoint.com/teams/MGI_FBS_TeamAPAC/Shared%20Documents/General/01_Demo_Data_And_Reports/02_BIT_Demo_Data/05_Demultiplexing/case3_10r_6r

   Barcode file:
   ```
   barcode1      TGGGAACGCTCCGGTT
   barcode2      TAATCCTTATCATATT
   barcode3      CAATGCGTAGGTTCGG
   barcode4      ATCCGCTCAGTAGGAG
   barcode5      GTCTCTAATCACCTAG
   ```

   Note: the red one stands for the 2nd barcode

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Parameters:

```
./bin/splitBarcode -B case3.barcode -1 case3_10r_6r_1.fq.gz -2 case3_10r_6r_2.fq.gz -b 200
10 1 -b 210 6 1 -r -o ./
```

4. Dual (10bp+10bp) barcode, only the first one is useable. Reversed.

   In this case, the first barcode will be used for the demultiplexing, while the second will be marked as not available (N's) and removed.

   Test file:
   https://bgitech.sharepoint.com/teams/MGI_FBS_TeamAPAC/Shared%20Documents/General/01_Demo_Data_And_Reports/02_BIT_Demo_Data/05_Demultiplexing/case4_10r_10n

   Barcode file:

```
barcode1        TGGGAACGCTNNNNNNNNNN
barcode2        TAATCCTTATNNNNNNNNNN
barcode3        CAATGCGTAGNNNNNNNNNN
barcode4        ATCCGCTCAGNNNNNNNNNN
barcode5        GTCTCTAATCNNNNNNNNNN
```

Note: the red one stands for the 2nd barcode

   Parameters:

```
./bin/splitBarcode -B case4.barcode -1 case4_10r_10n_1.fq.gz -2 case4_10r_10n_2.fq.gz -b
200 10 1 -b 210 10 1 -r -o ./
```

5. Dual (10bp+10bp) barcode, only the second one is useable. Reversed.

   In this case, the second barcode will be used for the demultiplexing, while the first will be marked as not available (N's) and removed.

   Test file:
   https://bgitech.sharepoint.com/teams/MGI_FBS_TeamAPAC/Shared%20Documents/General/01_Demo_Data_And_Reports/02_BIT_Demo_Data/05_Demultiplexing/case5_10n_10r

   Barcode file:

```
barcode1        NNNNNNNNNNCCGAGACGTT
barcode2        NNNNNNNNNNCATTCCTATT
barcode3        NNNNNNNNNNGTAGCGTCGG
barcode4        NNNNNNNNNNTTCGACCTAG
barcode5        NNNNNNNNNNCCTGTAGGAG
```

Note: the red one stands for the 2nd barcode

   Parameters:

```
./bin/splitBarcode -B case5.barcode -1 case5_10n_10r_1.fq.gz -2 case5_10n_10r_2.fq.gz -b
200 10 1 -b 210 10 1 -r -o ./
```

6. Dual (10bp+10bp) barcode for PE100 reads. reversed + forward.
   Test file:
   https://bgitech.sharepoint.com/teams/MGI_FBS_TeamAPAC/Shared%20Documents/General/01_Demo_Data_And_Reports/02_BIT_Demo_Data/05_Demultiplexing/case6_10r_10f_Dual

   Barcode file:

```
barcode1        TGGGAACGCTCCGAGACGTT
barcode2        TAATCCTTATCATTCCTATT
barcode3        CAATGCGTAGGTAGCGTCGG
```

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```
barcode4        ATCCGCTCAGCCTGTAGGAG
barcode5        GTCTCTAATCTTCGACCTAG
```

Note: the red one stands for the 2nd barcode

Parameters:

```
./bin/splitBarcode -B case6.barcode -1 case6_10r_10f_1.fq.gz -2 case6_10r_10f_2.fq.gz -b
200 10 1 -b 210 10 1 -r -o ./
```

Note: The 2 barcodes are not all reversed, so the analyst should only reverse 1 barcode in the preparation. For example, the barcode region in read2 is **AGCGTTCCCAAACGTCTCGG.** And the original barcodes are **TGGGAACGCT** and **AACGTCTCGG**. Since the parameter '-r' is applied to both barcodes, we need to reverse the second one, so the barcodes in file *case2.barcode* should be **TGGGAACGCTCCGAGACGTT.**

7. Dual (10bp+10bp) barcode + Single(10bp) barcode. Reversed.

   In this case, for single barcode, the second barcode will be used for the demultiplexing, while the first will be marked as not available (N's) and removed. All barcode pairs should be used for dual barcode.

   Test file:
   https://bgitech.sharepoint.com/teams/MGI_FBS_TeamAPAC/Shared%20Documents/General/01_Demo_Data_And_Reports/02_BIT_Demo_Data/05_Demultiplexing/case7_10n_10r_singleAndDual

   Barcode file:

```
barcode1        TGGGAACGCTCCGAGACGTT
barcode2        TAATCCTTATCATTCCTATT
barcode3        CAATGCGTAGGTAGCGTCGG
barcode4        ATCCGCTCAGCCTGTAGGAG
barcode5        GTCTCTAATCTTCGACCTAG
barcode6        GACGCCGGATNNNNNNNNNN
barcode7        GACGCATAATNNNNNNNNNN
barcode8        GACGTAAGAGNNNNNNNNNN
barcode9        GACGGTTCAGNNNNNNNNNN
barcode10       GACGACCTAANNNNNNNNNN
```

Note: the red one stands for the 2nd barcode

Parameters:

```
./bin/splitBarcode -B case7.barcode -1 case7_10n_10r_singleAndDual_1.fq.gz -2
case7_10n_10r_singleAndDual_2.fq.gz -b 200 10 1 -b 210 10 1 -r -o ./
```

8. Dual (10bp+10bp) barcode + Single(6bp) barcode. Reversed.

   In this case, the length for single barcode is not equal to half of the dual barcode, so they may not be analyzed as a batch. The analyst should use a 'twice-demultiplexing' strategy to analysis the dual barcode first, followed by the splitting by single barcode on the un-decoded reads of the first step.

   Test file:
   https://bgitech.sharepoint.com/teams/MGI_FBS_TeamAPAC/Shared%20Documents/General/01_Demo_Data_And_Reports/02_BIT_Demo_Data/05_Demultiplexing/case8_dual10_single6

   Barcode file1:

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```
barcode1       TGGGAACGCTCCGAGACGTT
barcode2       TAATCCTTATCATTCCTATT
barcode3       CAATGCGTAGGTAGCGTCGG
barcode4       ATCCGCTCAGCCTGTAGGAG
barcode5       GTCTCTAATCTTCGACCTAG
```

Barcode File2:

```
barcode6       CCGGATNNNNNNNNNNNNNN
barcode7       CATAATNNNNNNNNNNNNNN
barcode8       TAAGAGNNNNNNNNNNNNNN
barcode9       GTTCAGNNNNNNNNNNNNNN
barcode10      ACCTAANNNNNNNNNNNNNN
```

Note: the red one stands for the 2nd barcode

Parameters:

```
./bin/splitBarcode -B case8.dual.barcode -1 case8_dual10_single6_1.fq.gz -2
case8_dual10_single6_2.fq.gz -b 200 10 1 -b 210 10 1 -r -o ./dual_barcode
./bin/splitBarcode -B case8.single.barcode -1 dual_barcode/V350022662_L03_undecoded_1.fq.gz
-2 dual_barcode/V350022662_L03_undecoded_2.fq.gz -b 200 6 1 -b 206 14 1 -r -
o ./single_barcode
mkdir Final_Result
mv dual_barcode/*.fq.gz Final_Result/
mv single_barcode/*.fq.gz Final_Result/
```

----------END----------

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